Geometry
Unit 13
Lesson 3 Practice

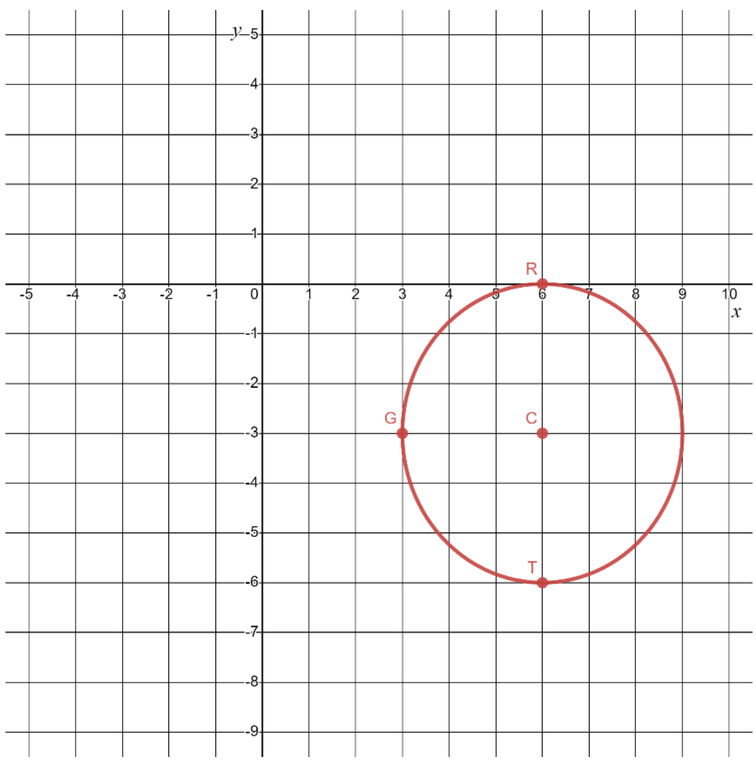
Name__Answer Key____
Date________________
Period_______________

The equation of circle C is $(x - 6)^2 + (y + 3)^2 = 9$.

1. Complete the table by substituting the given coordinates for x and y in the equation of circle C. Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$R(6,0)$	$(6-6)^2 + (0+3)^2 = 9$ $(0)^2 + (3)^2 = 9$	$9 = 9$
$G(3,-3)$	$(3-6)^2 + (-3+3)^2 = 9$ $(-3)^2 + (0)^2 = 9$	$9 = 9$
$T(6,-6)$	$(6-6)^2 + (-6+3)^2 = 9$ $(0)^2 + (-3)^2 = 9$	$9 = 9$

2. Graph circle C, $(x - 6)^2 + (y + 3)^2 = 9$.



3. Plot $R(6,0)$, $G(3,-3)$, $T(6,-6)$ on the circle.

4. Determine the length of the diameter on circle C.
Diameter $= 6$

Circle D has a radius of 2 with points $L(-3,7)$, $F(-1,5)$, and $T(1,7)$ on the circle.

5. Determine which two points are the endpoints of a diameter of circle D.
L and T

6. Find the center of circle D.
$(-1,7)$

7. Write the equation for circle D.
$(x + 1)^2 + (y - 7)^2 = 4$

8. The equation of circle W is $(x + 7)^2 + y^2 = 4$. Determine whether the following points lie inside, outside, or on circle W.

Point	Work	Location
$C(-7,2)$	$(-7+7)^2 + (2)^2 = 4$ $(0)^2 + 4 = 4$ $4 = 4$	o Inside circle W o Outside circle W ✖ On circle W
$N(-6,1)$	$(-6+7)^2 + (1)^2 = 4$ $1 + 1 = 4$ $2 = 4$	✖ Inside circle W o Outside circle W o On circle W
$T(-7,5)$	$(-7+7)^2 + (5)^2 = 4$ $(0)^2 + 25 = 4$ $25 = 4$	o Inside circle W ✖ Outside circle W o On circle W

9. The equation of circle R is $(x - 4)^2 + (y + 2)^2 = 13$.

Select all points that lie on circle R.

✖ $(6,1)$ $(6-4)^2 + (1+2)^2 = 4 + 9 = 13$
o $(4,-2)$
✖ $(1,0)$ $(1-4)^2 + (0+2)^2 = 9 + 4 = 13$
o $(0,0)$
o $(-4,-2)$

10. Circle M and circle T intersect at point H.

Equation of circle M is $x^2 + (y - 2)^2 = 10$ $(-3)^2 + (1-2)^2 = 9 + 1 = 10$ ✓
Equation of circle T is $(x + 4)^2 + (y - 2)^2 = 2$ $(-3+4)^2 + (1-2)^2 = 1 + 1 = 2$ ✓

Which coordinate pair is the location of point H.

o $(-5,2)$
o $(0,0)$
o $(-3,0)$
✖ $(-3,1)$